Appendix: Artifact Evaluation

Artifact Evaluation (AE)

Paper ID: pap142

This workflow evaluates A1 (TempGNN and three U280 accelerator baselines) and A2 (results/result.csv and the paper-figure generator).

Artifact: 10.5281/zenodo.21417187; source: TempGNN-Program/TempGNN; tag: sc26-ae-pap142-v1.0.

I. Download and Setup

```
git clone https://github.com/TempGNN-Program/
  ↪ TempGNN.git
cd TempGNN
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
```

```
Board: AMD/Xilinx Alveo U280,
  ↪ xcu280-fsvh2892-2L-e
Platform: xilinx_u280_gen3x16_xdma_1_202211_1
OS: Ubuntu 22.04 x86_64
XRT: 2.16.204
Vitis/Vivado: 2023.2 (xclbin rebuild only)
GCC: 11.4.0
GNU Make: 4.3
Python: 3.10 or newer
```

The U280 account and login instructions are delivered through the private SC26 submission channel. No credentials are stored in the artifact.

II. Review Plan

TABLE I
EXPECTED DIRECT REVIEW TIME.

Artifact	Setup	Run	Analysis	Expected result
A1	10–20	30–90	<5	Four fresh latency results; TempGNN lowest at about 1.3 ms
A2	<5	<1	2–5	Nine CSV/SVG pairs from results/result.csv

All times are minutes. Optional xclbin rebuilding is excluded from the direct review path.

III. Execute A1: U280 Latency

```
source /opt/xilinx/xrt/setup.sh
xrt-smi examine
make ae-core-u280 U280_CORE_DEVICE=0
  ↪ U280_CORE_REPETITIONS=3
```

The command performs:

1) A1-T1: check the four packaged U280 artifacts;
2) A1-T2: load six datasets and four models;
3) A1-T3: run TempGNN, MATG, ViTeGNN, and RTGA three times;
4) A1-T4: write per-implementation raw measurement CSVs; and
5) A1-T5: aggregate latency and update the AE report.

```
Datasets: WK, MC, RT, LM, WT, GT
Models: JODIE, TGN, TGAT, APAN
Repetitions: 3
```

Fresh results are written to results/reviewer_u280_runs/<run-id>/.

TABLE II
EXPECTED U280 MEAN LATENCY.

Implementation	Latency (ms)
TempGNN	1.296553
MATG	9.966618
ViTeGNN	2.869752
RTGA	4.192824

IV. Analyze A1

The run is successful when:

1) the command exits with status zero;
2) fresh rows exist for all four implementations;
3) all six datasets and four models are present; and
4) the aggregate latency follows the expected U280 behavior in Table II.

The principal behavior is that TempGNN has the lowest mean latency and remains close to 1.3 ms. This supports the target-centric execution, DDTC/OATS mechanisms, and U280 accelerator comparison in C1–C4.

V. Execute and Analyze A2

```
python3 -m scripts.reproduce_paper_figures
```

The command performs:

1) A2-T1: read results/result.csv;
2) A2-T2: validate and group rows by paper figure;
3) A2-T3: generate a CSV and SVG for each figure; and
4) A2-T4: write figure_data_manifest.csv.

Expected output under results/paper_reproduction/:

- Fig.2, Fig.4(a), and Fig.9(b);
- Fig.10, Fig.11, Fig.12, and Fig.13; and
- Fig.14(a), Fig.14(b), and the figure-data manifest.

The evaluation is successful when all nine CSV/SVG pairs exist and the manifest records results/result.csv as their input. These outputs provide the paper-result views associated with C1–C4.

VI. Baseline Status

- MATG: independently reproduced from the IPDPS 2022 paper and its partial public HLS headers; includes a distinct source, runner, and U280 xclbin.
- ViTeGNN: independently reproduced from the TPDS 2025 paper; includes a distinct source, runner, and U280 xclbin.
- RTGA: independently reproduced from the DAC 2024 paper; includes a distinct source, runner, and U280 xclbin.
- Cascade and TGLite-CPU: retained only as packaged paper comparison records; no fresh execution of those external stacks is claimed.
- TempGNN: includes HLS kernels, testbenches, Vitis build scripts, XRT host code, U280 board logs, and its own U280 xclbin.

The three baselines are independent, paper-based forward-path reproductions, not the authors' complete original stacks.

VII. Metric Meaning

DDTC is dependency-driven TDP construction. OATS is overlap-aware TDP synchronization.

packet_reuse_factor measures how many per-target packet materializations collapse into unique PHLE packets. memory_bytes is packet state/metadata traffic. cycles is the kernel's cycle proxy used by C-sim and exported fixture stats. HLS reports provide C/RTL latency after synthesis/cosim.

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